

Carina Zhang

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EDUCATION

University of Toronto

Bachelor of Science in Data Science and Computer Science; GPA: 4.0/4.0

Ontario, Canada

Sep. 2025 – Expected Jun. 2029

TECHNICAL SKILLS

Languages: Python, SQL, R, TypeScript/JavaScript

Backend & AI Agents: FastAPI, SQLAlchemy, Alembic, LangGraph, MCP, RAG, Celery, REST APIs, SSE

Databases & Storage: PostgreSQL, pgvector, Redis, SQLite, MinIO, AWS S3

DevOps & Tools: Docker, Docker Compose, Render, Git, GitHub, pytest

ML & Data: Scikit-learn, LightGBM, CatBoost, Pandas, NumPy, SciPy, Numba

PROJECTS

RBTLove - AI Companion | *Python, FastAPI, PostgreSQL, pgvector, LangGraph, MCP* Sep. 2026 – Present

- Led backend development in a 2-person team, designing a long-term memory layer and photo album service with 8 new REST endpoints, deployed as 6 Dockerized services (Postgres, Redis, MinIO, API, Celery worker, web).
- Built an **Agentic RAG** memory system behind an **MCP server** with 4 agent-callable tools, letting the LLM agent decide when to search or save memories instead of forcing a lookup every turn.
- Designed a 7-node **LangGraph** extraction pipeline, run as a **Celery/Redis** background job, that distills dialogue into memories and routes each candidate to insert, update, or discard in one graph run.
- Implemented **pgvector** semantic retrieval over a top-30 candidate pool, reranked by similarity, 14-day recency decay, and importance, with a shared 0.90 cosine threshold deduplicating both write paths.
- Migrated storage from SQLite to **PostgreSQL** using **SQLAlchemy** and **Alembic** (7 migrations), designing JSONB columns, an IVFFlat vector index, and a non-destructive version chain; covered by 91 pytest tests.

Wellbore Geology Prediction | *Python, FastAPI, Docker, AWS S3, LightGBM, CatBoost* May 2026 – Aug. 2026

- Built and deployed an end-to-end ML system predicting true vertical thickness (TVT) from horizontal well logs, ranking 1212/6191 (top 20%) on Kaggle with a stacked LightGBM/CatBoost ensemble.
- Engineered 150+ features from raw well logs, including gamma-ray alignment signals from 4 algorithms: a Particle Filter, beam-search sequence aligner, multi-scale NCC, and Sakoe-Chiba-constrained DTW.
- Accelerated feature generation 10× via Numba JIT, cutting full-dataset runtime from hours to minutes.
- Trained 6 base models (3 LightGBM, 3 CatBoost) under 5-fold GroupKFold CV to prevent well-level leakage; a RidgeCV stacking meta-learner beat the best single model on RMSE.
- Designed a KNN formation-plane estimator and cKDTree-based ANCC interpolator to impute missing geology from neighboring wells, increasing feature coverage across wellbore datasets.
- Served the ensemble as a **FastAPI** microservice with thread-pooled inference across 2 endpoints (single- and batch-well), enabling on-demand TVT predictions without retraining.
- Deployed with **Docker** on Render using 2 config-driven model profiles (full ensemble vs. lightweight demo), with model artifacts fetched on demand from **AWS S3**.

EXPERIENCE

New Oriental Education and Technology Group

Teaching Assistant

Shenzhen, China

Jun. 2025 – Aug. 2025

- Facilitated classroom activities and behavior management for 20+ students across daily sessions.
- Built an Excel tracking system for 20+ students' scores across multiple assessments, enabling instructors to monitor progress and flag at-risk students in real time.